PHYCSS FORM ONE

4. (i) B (ii) A (iii) A (iv) A (v) B
(vi) D (vii) C (viii) B (ix) A (x) B
1 mark ⊛ = 10 marks

2 (i) B (ii) C (iii) D (iv) A (v) E
1 Mark ⊛ = 10 marks

3 (i) Physical quantity
(ii) Technology
(iii) sinking
(iv) Parallax error
(v) Relative density.
2 marks ⊛ = 10 marks

4(a) (i) Archimedes principle state that when a body is partially or totally immersed in a fluid it experiences an upthrust which is equal to the weight of the fluid displaced by the body. 2 marks

(ii) Law of floatation state that ~~when~~ ~~a body~~ A floating body displaces its own weight of the fluid in which it floats
2 marks

4(b) solution

Data given

Weight in air = 0.8N − W_1

Apparent weight = 0.5N − W_2

(i) Upthrust = Weight in air − Apparent wt 1marks

= 0.8N − 0.5N ½ mark

= 0.3N ½ marks

(ii) $R.D = \frac{W_1}{W_1 - W_2}$ ½ mark

$= \frac{0.8N}{0.8N - 0.5N}$ ½ mark

$= \frac{0.8N \times 10}{0.3N \times 10}$ ½ mark

$= \frac{8}{3}$

R.D = 2.67 ½ mark

(iii) Density = R.D x Density of water 1marks

= 2.67 x 1g/cm³ ½ mark

= 2.67g/cm³ ½ mark

or

= 2.67 x 1000kg/m³ ½ mark

= 2670 kg/m³ ½ mark

5 (a) Laboratory is the special room or building designed for conducting scientific experiments. 1mark

(ii) Laboratory apparatus is the special tools and instruments commonly used to carry out the experiments in the laboratory. 1mark

(b) (i) Laboratory should be well ventillated and his door opened outward.

(ii) Fire extinguishers should fitted in an accessible position with using instructions

(iii) Labaratory floor should not be polished to avoid slippery

(iv) First aid kit must be present in the Laboratory

(v) Cabinets and drawers must present for storing apparatus

(vi) All apparatus should have checked regularly to ensure they are safe for use.

(vii) Emergence exit should be present and easy to access and use.

(any 5) 1mark (b) = 5

5(c) (i) Metre ruler
(ii) Tape Measure
(iii) Micrometer screw gauge
(iv) Vernier calliper (any 3) 1 mark @ = 3 marks

6(a) Density is the mass of substance per unity volume While Relative density is the ratio of density of substance to the density of water. 2 marks

(b) (i) Used in design Various structure like Ship, Airplanes etc.
(ii) Used to determine density of Unknown substance
(iii) Used to select building materials
(iv) Used to design equipment used in swimming pools and diving.
(v) Used in identification of gemstones
(vi) Used by geologist and Mineralogist to help determine the Mineral content of a rock or other sample. (any 5) 1 mark @ = 5 marks

6(c) solution:

Data given

Mass of empty bottle = 24.2g – M_0

Mass of bottle + Content = 67.81g – M_1

Volume of Liquid = 50ml ($50cm^3$).

since $1ml = 1cm^3$

Mass of turpentine = $M_1 - M_0$ ½ marks

= 67.81g – 24.2g ½ marks

= 43.61g · ½ marks

Density = $\frac{\text{Mass}}{\text{Volume}}$ ½ mark

= $\frac{43.61g}{50cm^3}$ ½ marks

$0.87g/cm^3$ · ½ marks

07. (a) (i) Cohesion is the force of attraction between molecules of the same materials.

00 $\frac{1}{2}$ mark.

ii) Osmosis is the movement of solvent from region of low concentration to region of high concentration through semi-permeable membrane.

00 $\frac{1}{2}$ marks

iii) Surface tension is the ability of surface of liquid to behave like a fully stretched elastic skin.

00 $\frac{1}{2}$ marks

b) Factors affecting surface tension:

i) Nature of liquid

ii) Contamination

iii) Temperature.

00 $\frac{1}{2}$ @ = 01 $\frac{1}{2}$

c) i) Hooke's law states that "Within elastic limit extension is directly proportional to applied force.

02 marks

ii) Mass	Weight
- Is quantity of matter in object	- Is force of gravity on object 01 @
- It is constant	- Varies with position = 05 marks
- Measured by using beam balance	- Measured by Spring balance
- SI Unit is Kg	- SI Unit Newton (N)
- It is fundamental quantity	- It's derived quantity.

8(a) (i) Mass is the quantity of matter an object contains.

(ii) Volume is amount of space an object occupies.

(iii) Weight is the force of gravity on an object.

(b) Fundamental quanty is the quantity that can not obtained from any other physical quantity. While Derived quantity is the quantity that can be obtained from other physical quantities by Multiplication, or division 2 marks

Examples of fundamental quantity

(i) Time, (ii) Length (iii) mass (iv) Electric current (v) Luminous intensity (vi) Amount of substance (any 2.) ½ mark @ = 1 mark

Examples of Derived quantities

(i) Area (ii) Volume (iii) Weight (iv) Work (v) Velocity (any 2). ½ mark @

Total 1 marks

8c

Quantity	SI unit	symbol
Area	Metre squared ½ marks	M^2 ½ mark
Volume	Metre cubed ½ marks	M^3 ½ mark
Weight	Newton ½ mark	N ½ mark
Luminous intensity	Candela ½ mark	cd ½ mark
Amount of substance	Mole ½ mark	Mol ½ mark
Electric current	Ampere ½ mark	A ½ mark

½ mark @ = 6 marks

9(a) Error is a ~~deviation~~ deviation from the true reading
or
Error is a Measure of estimated difference between the observed and true value of a quantity that is being Measured or observed
1 mark @ = 1 mark

(b) (i) Parallax error
(ii) Zero error
(iii) Instrumental error
1 mark @ = 3 marks

(c) (i) Fault during Manufacture
(ii) Damage during use
(iii) Poor Storage
(iv) Human factor
1 mark @ = 4 marks

9(d) Weight = 600 N
gravitation for on the moon = $\frac{1}{6}$

Weight of Zawadi on the moon = $60N \times \frac{1}{6}$ 1mark
$= 10N$. 1marks
2marks

10(a) (i) Fill the graduated cylinder with water to certain level.
(ii) carefully Measure the initial volume of water V_1
(iii) Gently Lower irregular body (stone) into the water
(iv) Measure the final volume of water (V_2)
(v) The difference between final volume and the initial volume gives the volume of the Stone
$V = V_2 - V_1$.
1mark @ = 5marks

(b) solution
Data given
Mass of empty beaker = 48g – Mo
Mass of beaker + Liquid = 60g – M1
(i) Mass of Liquid = $M_1 - M_0$ $\frac{1}{2}$ marks
$= 60g - 48g$ $\frac{1}{2}$ marks
$= 12g$. $\frac{1}{2}$ marks

10(b)ii Density = $\frac{\text{mass of Liquid}}{\text{Volume}}$ $\frac{1}{2}$ mark

$= \frac{12g}{25cm^3}$ 1 marks

$0.48 g/cm^3$ $\frac{1}{2}$ marks

(iii) R.D = $\frac{\text{Density of substance}}{\text{Density of water}}$ $\frac{1}{2}$ mark

$\frac{0.48 g/cm^3}{1 g/cm^3}$ $\frac{1}{2}$ mark

R.D = 0.48 $\frac{1}{2}$ marks